GREEEN tallgrass prairie restoration: RLQ analyses (Lux Arbor)

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Contents

Preparation **1**
Checks . . . 2
Finalize preparation . . . 3

Statistics **3**
Descriptive analyses . . . 3
Ordinations . . . 4
RLQ analyses . . . 4
Global test . . . 4
Traits by environment crossed table . . . 4

Plots **5**

Save **9**

Session info **9**

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Preparation

```
library(here)
library(tidyverse)
library(ade4)
```

Packages

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types =
    cols(
    .default = "?",
    seeding_time = "f"
  )) %>%
```

```r
  arrange(id_plot_year) %>%
  filter(
    site == "Lux Arbor",
    richness_type == "seeded_richness",
    treatment_id %in% c("1", "2", "3", "4")
  ) %>%
  mutate(
    treatment = str_c(seeding_time, herbicide, seeded_pool, sep = "_"),
    treatment = factor(treatment)
  ) %>%
  filter(treatment != "unseeded_0_0") %>%
  select(
    id_plot_year, year, treatment, water_cap, cover_non_seeded,
    cover_seeded_grass
    )

traits <- read_csv(
  here("data", "processed", "data_processed_traits.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(.default = "?")
  ) %>%
  filter(seeded == "1", family != "Poaceae") %>%
  select(accepted_name, sla, height, seedmass) %>%
  mutate(seedmass = log(seedmass)) %>%
  drop_na()

species <- read_csv(
  here("data", "processed", "data_processed_species.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(.default = "?")
  ) %>%
  filter(plot_size %in% c("1", "25")) %>%
  mutate(abundance = 1) %>%
  select(id_plot_year, accepted_name, abundance) %>%
  semi_join(sites, by = "id_plot_year") %>%
  semi_join(traits, by = "accepted_name") %>%
  arrange(id_plot_year)
```

Load data

Checks

```r
species %>%
  group_by(id_plot_year) %>%
  count()
```

Seeded species richness

```
## # A tibble: 104 x 2
## # Groups:   id_plot_year [104]
##    id_plot_year     n
##    <chr>        <int>
##  1 Lux_12_2015     10
##  2 Lux_12_2016     10
##  3 Lux_12_2017     10
##  4 Lux_12_2018      9
```

```
##  5 Lux_12_2019      8
##  6 Lux_12_2025      4
##  7 Lux_17_2015      3
##  8 Lux_17_2016      5
##  9 Lux_17_2017      9
## 10 Lux_17_2018      8
## # i 94 more rows
```

```
traits %>% anti_join(species, by = "accepted_name")
```

Check non-occuring seeded species

```
## # A tibble: 6 x 4
##   accepted_name                  sla height seedmass
##   <chr>                        <dbl>  <dbl>    <dbl>
## 1 Dalea purpurea                191.  0.91     -6.07
## 2 Liatris cylindracea           110.  0.564    -5.73
## 3 Lupinus perennis              167.  0.326    -3.84
## 4 Silphium terebinthinaceum     171.  0.716    -4.53
## 5 Solidago juncea               130.  0.783    -7.79
## 6 Symphyotrichum oolentangiense 143.  0.362    -8.68
```

Finalize preparation

```
traits <- traits %>%
  semi_join(species, by = "accepted_name") %>%
  column_to_rownames(var = "accepted_name")

sites <- sites %>%
  semi_join(species, by = "id_plot_year") %>%
  column_to_rownames(var = "id_plot_year")

species <- species %>%
  pivot_wider(
    names_from = "accepted_name", values_from = "abundance", values_fill = 0
    ) %>%
  column_to_rownames(var = "id_plot_year")
```

Statistics

Descriptive analyses

```
sites %>%
  group_by(year) %>%
  count()
```

```
## # A tibble: 6 x 2
## # Groups:   year [6]
##    year     n
##   <dbl> <int>
## 1  2015    14
## 2  2016    18
## 3  2017    18
```

```
## 4  2018    18
## 5  2019    18
## 6  2025    18
```

Ordinations

```
dudi_l <- dudi.coa(species, scannf = FALSE)
dudi_r <- dudi.hillsmith(
  sites,
  row.w = dudi_l$lw,
  scannf = FALSE
  )
dudi_q <- dudi.pca(
  traits,
  row.w = dudi_l$cw,
  scannf = FALSE
  )
```

RLQ analyses

```
m1 <- rlq(
  dudiR = dudi_r,
  dudiL = dudi_l,
  dudiQ = dudi_q,
  scannf = FALSE
  )
```

Global test

```
randtest(m1, nrepet = 999, modeltype = 6)
```

```
## class: krandtest lightkrandtest
## Monte-Carlo tests
## Call: randtest.rlq(xtest = m1, nrepet = 999, modeltype = 6)
##
## Number of tests:   2
##
## Adjustment method for multiple comparisons:   none
## Permutation number:   999
##      Test       Obs   Std.Obs   Alter Pvalue
## 1 Model 2 0.1291485 15.557415 greater  0.001
## 2 Model 4 0.1291485  2.067795 greater  0.042
```

Traits by environment crossed table

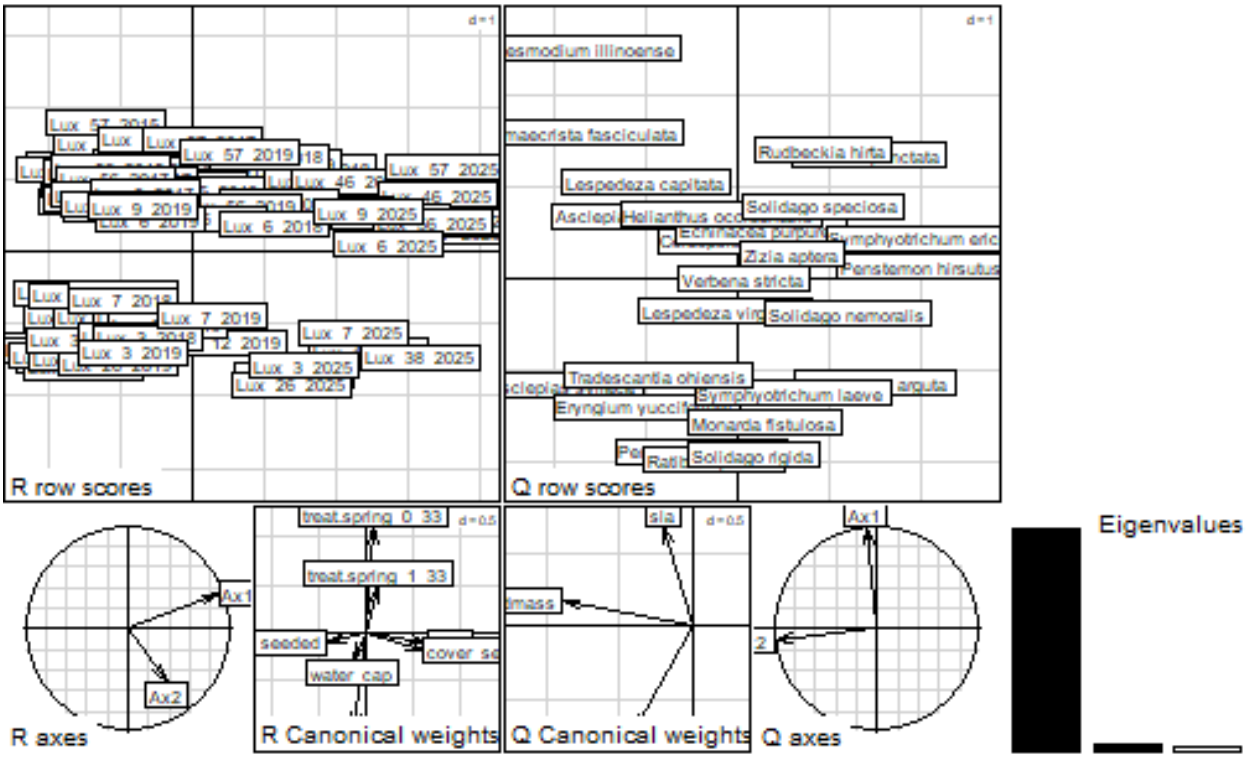
```
m1$tab
```

```
##                              sla      height    seedmass
## year               -0.0553359388 -0.09100603 -0.20077237
## treat.fall_0_33    -0.0352059036  0.06892163  0.03075780
## treat.spring_0_33  -0.0007545037 -0.09902116  0.01473937
## treat.spring_1_33   0.0469446306 -0.01186907 -0.05222636
## water_cap          -0.0212619477  0.01978769  0.05335722
```

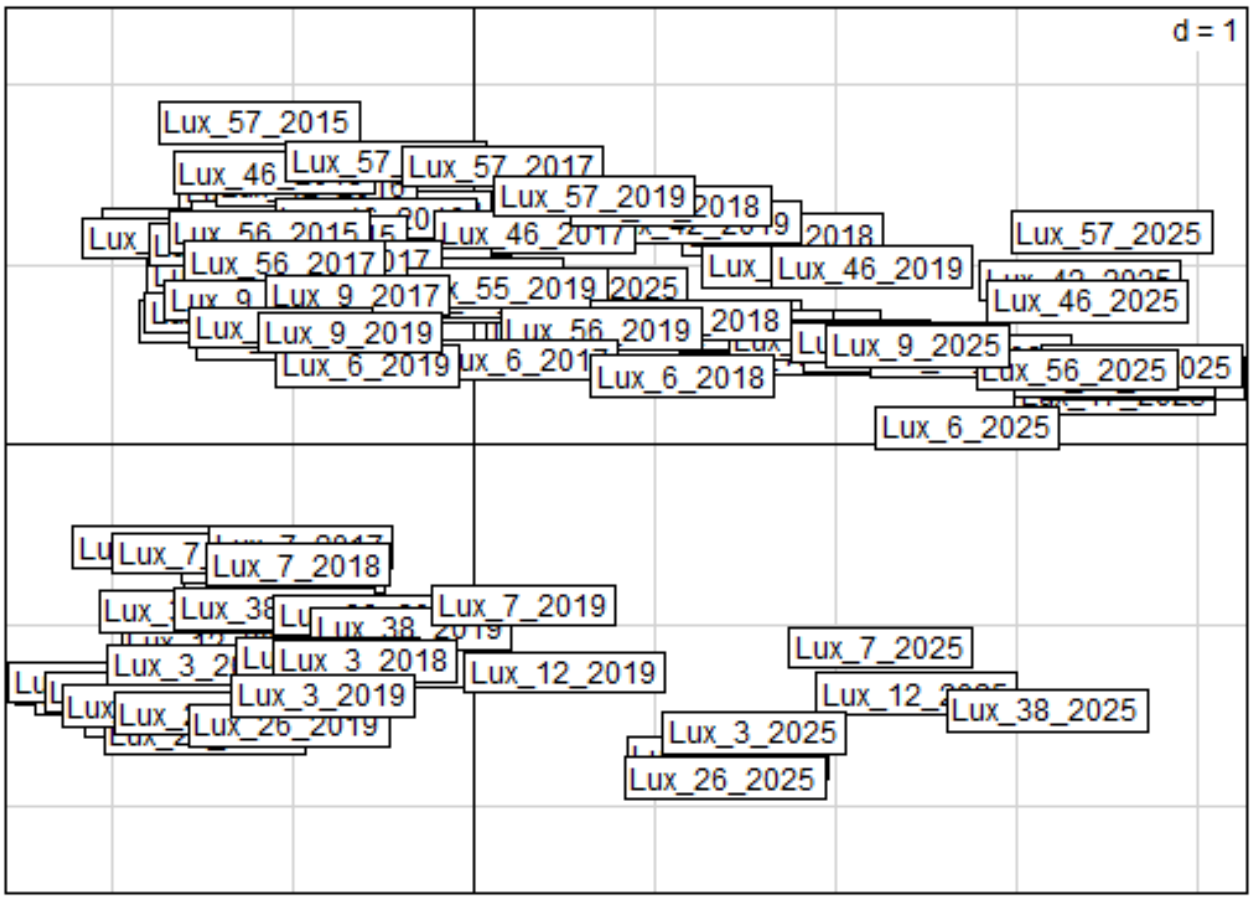
```
## cover_non_seeded    0.0171048445  0.05755098  0.13280700
## cover_seeded_grass -0.0577130707 -0.08164154 -0.18825188
```

Plots

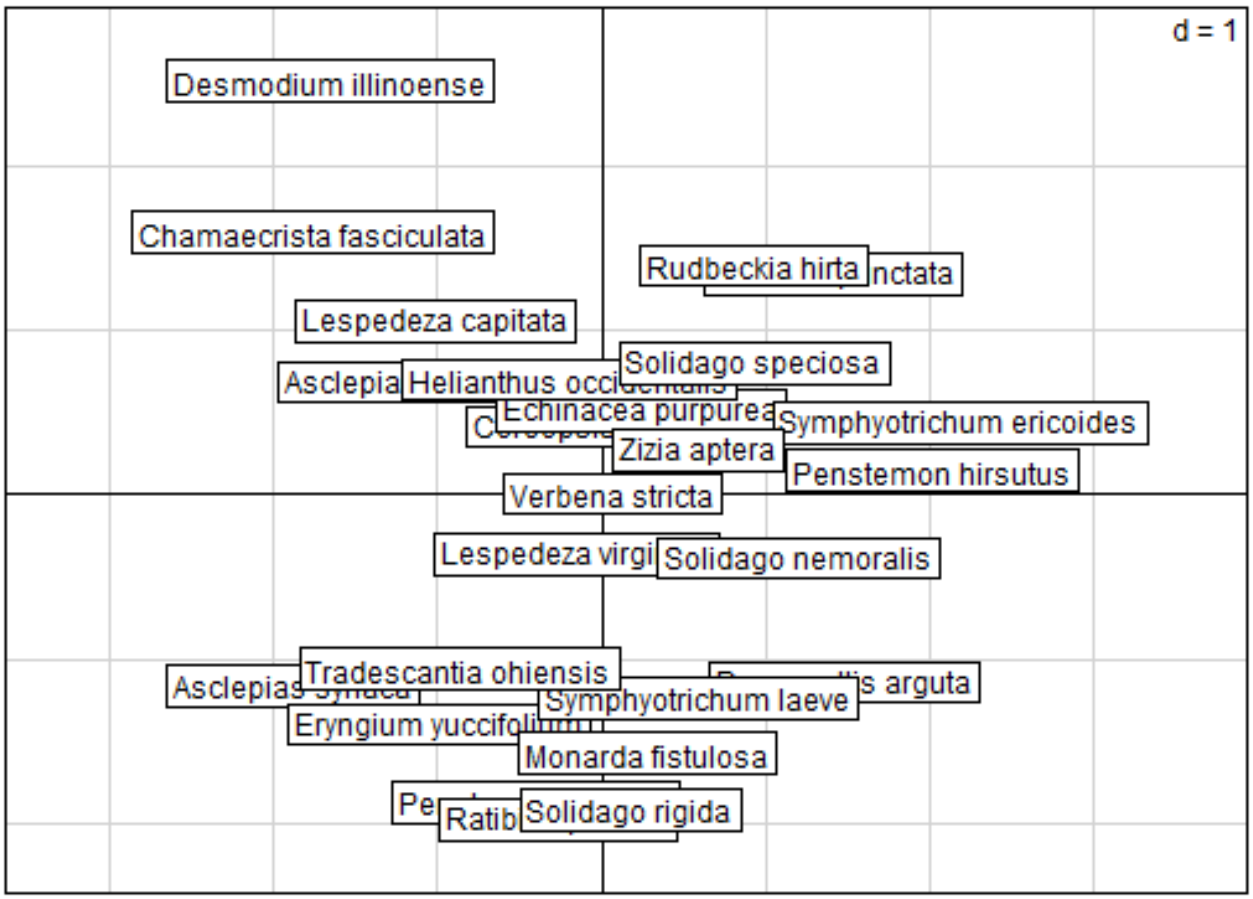
```
plot(m1)
```



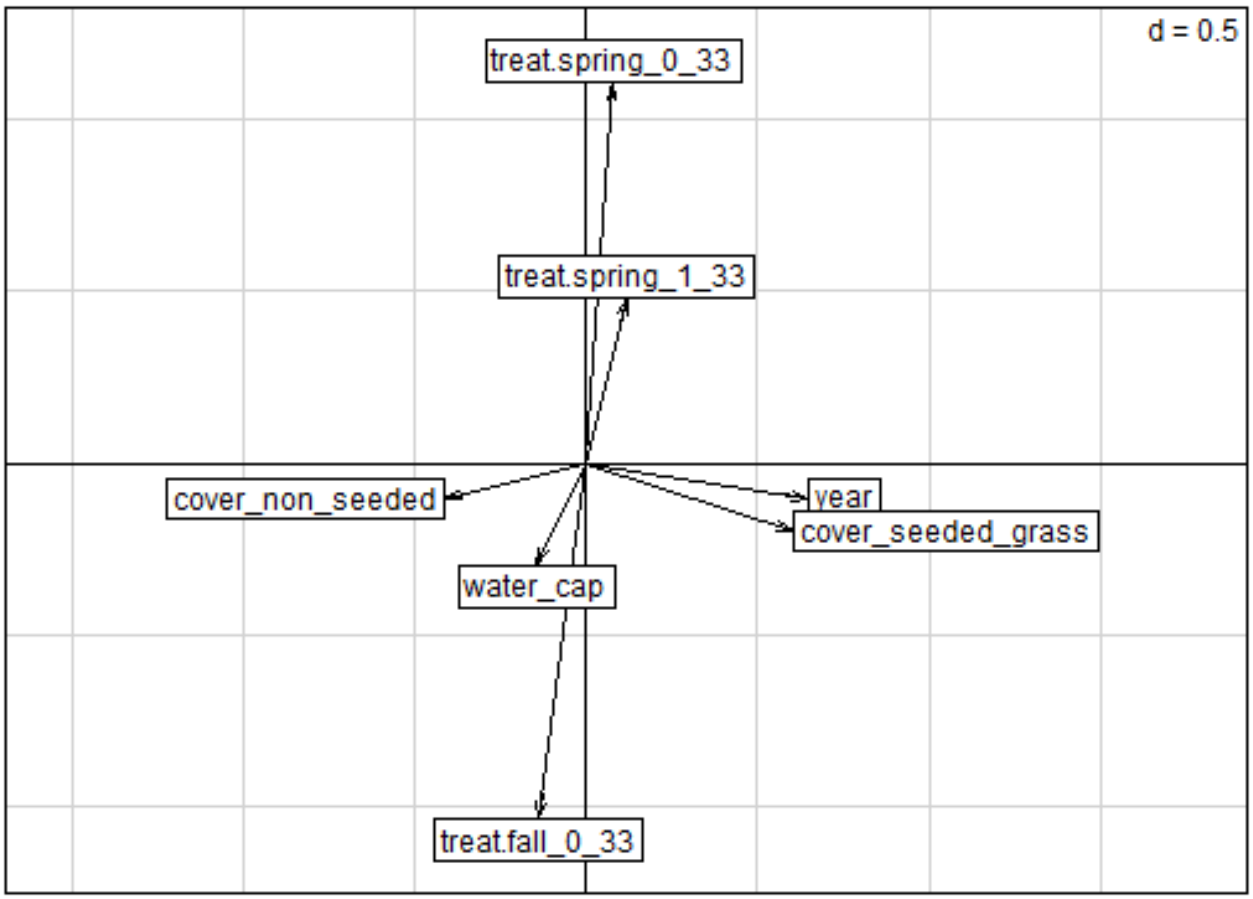

```
s.label(m1$lR)
```


d = 1
Lux_57_2015
Lux_57_2017
Lux_57_2019
Lux_46_2017
Lux_46_2019
Lux_57_2025
Lux_46_2025
Lux_9_2017
Lux_9_2019
Lux_9_2025
Lux_56_2019
Lux_56_2025
Lux_6_2019
Lux_6_2018
Lux_6_2025
Lux_7_2018
Lux_7_2019
Lux_7_2025
Lux_3_2018
Lux_3_2019
Lux_12_2019
Lux_38_2025
Lux_3_2025
Lux_26_2025

```
s.label(m1$lQ)
```


d = 1
Desmodium illinoense
Chamaecrista fasciculata
Rudbeckia hirta
Lespedeza capitata
Solidago speciosa
Asclepia
Helianthus occi
Echinacea purpurea
Symphyotrichum ericoides
Zizia aptera
Penstemon hirsutus
Verbena stricta
Lespedeza virgi
Solidago nemoralis
Tradescantia ohiensis
Asclepias
Symphyotrichum laeve
Eryngium yuccifo
Monarda fistulosa
Solidago rigida

```
s.arrow(m1$l1)
```


d = 0.5
treat.spring_0_33
treat.spring_1_33
cover_non_seeded
year
cover_seeded_grass
water_cap
treat.fall_0_33

```
s.arrow(m1$c1)
```

Save

Session info

```
## R version 4.5.0 (2025-04-11 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] ade4_1.7-23     lubridate_1.9.4 forcats_1.0.0   stringr_1.5.1
```

```
##  [5] dplyr_1.1.4     purrr_1.1.0     readr_2.1.5     tidyr_1.3.1
##  [9] tibble_3.3.0    ggplot2_3.5.2   tidyverse_2.0.0 here_1.0.1
## [13] tinytex_0.57
##
## loaded via a namespace (and not attached):
##  [1] bit_4.6.0          gtable_0.3.6       crayon_1.5.3       compiler_4.5.0
##  [5] Rcpp_1.1.0         tidyselect_1.2.1   parallel_4.5.0     scales_1.4.0
##  [9] yaml_2.3.10        fastmap_1.2.0      R6_2.6.1           generics_0.1.4
## [13] knitr_1.50         MASS_7.3-65        rprojroot_2.0.4    pillar_1.11.0
## [17] RColorBrewer_1.1-3 tzdb_0.5.0         rlang_1.1.6        utf8_1.2.6
## [21] stringi_1.8.7      xfun_0.52          bit64_4.6.0-1      timechange_0.3.0
## [25] cli_3.6.5          withr_3.0.2        magrittr_2.0.3     digest_0.6.37
## [29] grid_4.5.0         vroom_1.6.5        rstudioapi_0.17.1  hms_1.1.3
## [33] lifecycle_1.0.4    vctrs_0.6.5        evaluate_1.0.4     glue_1.8.0
## [37] farver_2.1.2       rmarkdown_2.29     tools_4.5.0        pkgconfig_2.0.3
## [41] htmltools_0.5.8.1
```